

Problem Set 10 Answers

1. There are 48 equally likely outcomes. Thus:

$$P(X = 1) = 13/48$$

$$P(X = 2) = 11/48$$

$$P(X = 3) = 9/48 = 3/16$$

$$P(X = 4) = 7/48$$

$$P(X = 5) = 5/48$$

$$P(X = 6) = 3/48 = 1/16$$

We compute $E(X) = (1)(13/48) + (2)(11/48) + (3)(9/48) + (4)(7/48) + (5)(5/48) + (6)(3/48) = 133/48$.

2. To have exactly x green bears, there are $\binom{3}{x}$ ways that x out of the 3 bears will be green, so $P(X = x) = \binom{3}{x}(1/6)^x(5/6)^{3-x}$. In other words, we have $P(X = 0) = (5/6)^3$, $P(X = 1) = (3)(1/6)(5/6)^2$, $P(X = 2) = (3)(1/6)^2(5/6)$, $P(X = 3) = (1/6)^3$. Now we compute $E(X) = (0)(5/6)^3 + (1)(3)(1/6)(5/6)^2 + (2)(3)(1/6)^2(5/6) + (3)(1/6)^3 = 1/2$.

3. We have:

$$P(X = 1) = 3/10$$

$$P(X = 2) = (7/10)(3/9) = 7/30$$

$$P(X = 3) = (7/10)(6/9)(3/8) = 7/40$$

$$P(X = 4) = (7/10)(6/9)(5/8)(3/7) = 1/8$$

$$P(X = 5) = (7/10)(6/9)(5/8)(4/7)(3/6) = 1/12$$

$$P(X = 6) = (7/10)(6/9)(5/8)(4/7)(3/6)(3/5) = 1/20$$

$$P(X = 7) = (7/10)(6/9)(5/8)(4/7)(3/6)(2/5)(3/4) = 1/40$$

$$P(X = 8) = (7/10)(6/9)(5/8)(4/7)(3/6)(2/5)(1/4)(3/3) = 1/120$$

Now we compute $E(X) = (1)(3/10) + (2)(7/30) + (3)(7/40) + (4)(1/8) + (5)(1/12) + (6)(1/20) + (7)(1/40) + (8)(1/120) = 11/4$.

4. The probability mass function is $P(X = x) = \binom{5}{x}(1/3)^x(2/3)^{5-x}$ for $0 \leq x \leq 5$. So the expected value of X is $E(X) = (0)(2/3)^5 + (1)(5)(1/3)(2/3)^4 + (2)(10)(1/3)^2(2/3)^3 + (3)(10)(1/3)^3(2/3)^2 + (4)(5)(1/3)^4(2/3) + (5)(1/3)^5 = 5/3$.

5. The expected value of X is $E(X) = \sum_{x=0}^{\infty} (x) \left(\frac{3^x e^{-3}}{x!} \right)$. The $x = 0$ term is 0, so we can rewrite this as: $E(X) = \sum_{x=1}^{\infty} (x) \left(\frac{3^x e^{-3}}{x!} \right) = (3) \sum_{x=1}^{\infty} \frac{3^{x-1} e^{-3}}{(x-1)!} = (3) \sum_{x=0}^{\infty} \frac{3^x e^{-3}}{x!}$.

We know that the sum is 1 because this is the probability mass function of X (that was given in the problem statement), so we conclude that $E(X) = 3$.